CSM
Fall 2025

EECS 16B
Worksheet 6

1. Simply Complex Problems

(a) Let $z_1 = x_1 + iy_1 = r_1 e^{i\theta_1}$ and $z_2 = x_2 + iy_2 = r_2 e^{i\theta_2}$. Find the real part and imaginary part of the product $z_1 z_2$ and the quotient z_1/z_2 in terms of the Cartesian components x_1, x_2, y_1, and y_2. Then find the real and imaginary part of the product $z_1 z_2$ and the quotient z_1/z_2 in terms of the polar components r_1, r_2, θ_1, and θ_2.

(b) Let $z = x + iy = re^{i\theta}$. Find z^2 and $|z|^2$ first in terms of x and y and then in terms of r and θ. Under what circumstances does $|z|^2 = z^2$?

(c) Use Euler's formula on both sides of $e^{i(\alpha+\beta)} = e^{i\alpha}e^{i\beta}$ to derive the formulas for $\cos(\alpha+\beta)$ and $\sin(\alpha+\beta)$.

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(d) Let $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$. Show that

$$|z_1 + z_2| = \sqrt{r_1^2 + r_2^2 + 2r_1 r_2 \cos(\theta_1 - \theta_2)} \tag{1}$$

(e) Use this result to show the triangle inequality,

$$|z_1 + z_2| \leq |z_1| + |z_2| \tag{2}$$

Under what conditions is this an equality?

(f) Show De Moivre's formula,

$$(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta) \tag{3}$$

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(g) Use De Moivre's formula to show the double angle formulas:

$$\begin{cases} \cos 2\theta = \cos^2\theta - \sin^2\theta \\ \sin 2\theta = 2\cos\theta\sin\theta \end{cases} \tag{4}$$

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2. Phasor Analysis

Any sinusoidal time-varying function $v(t)$, representing a voltage or a current, can be expressed in the form

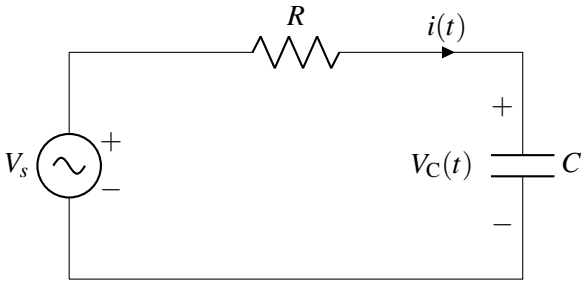
$$v(t) = \widetilde{V} e^{j\omega t} + \overline{\widetilde{V}} e^{-j\omega t}, \tag{5}$$

For a sinusoidal cosine wave $v(t) = V_0 \cos(\omega t + \phi)$, a phasor is defined as:

$$\widetilde{V} = \frac{V_0}{2} e^{j\phi} \text{ and its complex conjugate } \overline{\widetilde{V}} = \frac{V_0}{2} e^{-j\phi}$$

$\widetilde{V}$ is a time-independent function, and is referred to as the phasor representation of $v(t)$.

The phasor analysis method consists of five steps. Consider the RC circuit below.



The voltage source is given by

$$V_s(t) = 8\cos\left(\omega t - \frac{\pi}{4}\right), \tag{6}$$

with $\omega = 10^3$ rad/s, $R = 1$ kΩ, and $C = 1$ μF.

Our goal is to obtain a solution for $V_C(t)$ and $i(t)$ with the sinusoidal voltage source $V_s(t)$.

(a) **Step 1: Convert $v(t)$ as a sum of complex exponentials**

All voltages and currents with known sinusoidal functions should be expressed in the standard exponential format. **Convert $v_s(t)$ into an exponential and write down its phasor representation $\widetilde{V}_s$.**

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(b) **Step 2: Transform circuits to phasor domain**

The voltage source is represented by its phasor $\widetilde{V}_s$. The current $i(t)$ is related to its phasor counterpart $\widetilde{I}$. What are the phasor representations of R and C?

(c) **Step 3: Cast KCL and/or KVL equations in phasor domain**

Use Kirchhoff's laws to write down a loop equation that relates all phasors in Step 2.

(d) **Step 4: Solve for unknown variables**

Solve the equation you derived in Step 3 for $\widetilde{I}$ and $\widetilde{V}_c$. What is the polar form of $\widetilde{I}$ ($Ae^{i\theta}$, where A is a positive real number)?

(e) **Step 5: Transform solutions back to time domain**

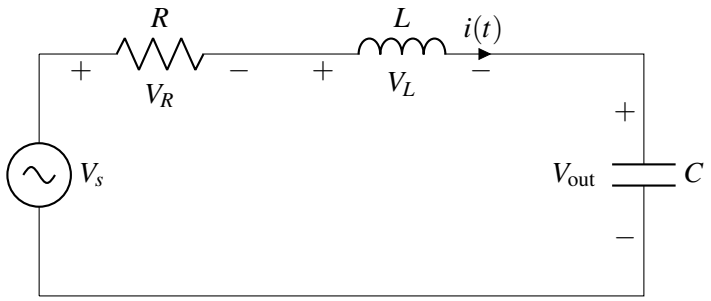
To return to time domain, we apply the fundamental relation between a sinusoidal function and its phasor counterpart. What is $i(t)$ and $v_c(t)$? What is the phase difference between $i(t)$ and $v_c(t)$?

It's important to keep in mind that the phasor analysis above only applies to sinusoidal inputs. Don't make the mistake of doing phasor analysis on DC inputs!

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3. RLC circuit Phasor Analysis

In this question, we will take a look at an electrical systems described by second order differential equations and analyze it using the phasor domain. Consider the circuit below where $R = 3\text{k}\Omega$, $L = 1\text{mH}$, $C = 100\text{nF}$, and $V_\text{s} = 5\cos\left(1000t + \frac{\pi}{4}\right)$:



(a) What are the impedances of the resistor, inductor and capacitor, Z_R, Z_L, and Z_C?

(b) Solve for $\widetilde{V}_\text{out}$ in phasor form.

(c) What is V_out in the time domain?

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(d) Solve for the current $i(t)$

(e) Solve for the transfer function $H(\omega) = \frac{\widetilde{V}_{\text{out}}}{\widetilde{V}_{\text{s}}}$
Leave your answer in terms of R, L, C, and ω.

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